***Solution:***



S.P of the bicycle = ₹4275

Loss = 5%

$$\therefore \text{ C.P} = \frac{100}{(100 - Loss\%)} \times \text{ S.P}$$

$$= \frac{100}{95} \times 4275 = ₹4500$$

Now, C.P = ₹4500 and the desired profit = 5%

$$\therefore \text{ Desired S.P } = \frac{(100 + Profit\%)}{100} \times \text{C.P}$$

$$= \frac{100+5}{100} \times 4500 = 105 \times 45 = ₹4725$$

∴ To have a profit of 5%, he should sell the bicycle for ₹4725.

**Example 4.10**

The price of a rain coat was slashed from ₹1060 to ₹901 by a shopkeeper in the rainy season to boost the sales. Find the rate of discount given by him.



***Solution:***

Discount = Marked Price –Selling Price

= 1060 – 901

= ₹159

$$\therefore \text{ Discount \%} = \left(\frac{159}{1060} \times 100\right)\%$$

= 15%



**Think**

A shopkeeper marks the price of a marker board 15% above the cost price and then allows a discount of 15% on the marked price. Does he gain or lose in the transaction?

**Example 4.11**

Find the single discount in percentage which is equivalent to two successive discounts of 25% and 20% given on an article.

***Solution:***

Let the marked price of an article be ₹100.

First discount of 25% $= 100 \times \frac{25}{100} = ₹25$

Price after first discount $= 100 - 25 = ₹75$

Second discount of 20% $= 75 \times \frac{20}{100} = ₹15$



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